

(1) Convergence/Divergence Pattern (P1)																											
Mathematical Definition (M1) Let $\bar{X}_A(t)$ and $\bar{X}_B(t)$ be two time series. Let $D(t) = \bar{X}_A(t) - \bar{X}_B(t) $ represent the distance between them at time t . • A Divergence Pattern exists if $D(t_{\text{end}}) > D(t_{\text{start}})$. • A Convergence Pattern exists if $D(t_{\text{end}}) < D(t_{\text{start}})$.		Textual Description (T1) This pattern identifies when two different timeseries are moving closer together or pulling further apart over a period of time.																									
Instances in Report (E1.text) “In terms of patent growth, patent families assigned to GANs show the strongest increase over the past decade. However, patent growth of GAN patent families has slowed down and GAN patent families have increased only moderately over the last three years. In contrast, diffusion models and LLMs show much higher growth rates over the last three years...”		Instances in Data (E1.tab) <table><tr><th>Grant Year</th><th>GAN Count</th><th>LLM/Diff Count</th></tr><tr><td>2018 (t_{start})</td><td>1,240</td><td>1,100</td></tr><tr><td>2019</td><td>1,350</td><td>1,450</td></tr><tr><td>2020</td><td>1,420</td><td>2,100</td></tr><tr><td>2021</td><td>1,480</td><td>4,200</td></tr><tr><td>2022</td><td>1,510</td><td>8,900</td></tr><tr><td>2023</td><td>1,550</td><td>14,300</td></tr><tr><td>2024 (t_{end})</td><td>1,590</td><td>22,100</td></tr></table>		Grant Year	GAN Count	LLM/Diff Count	2018 (t_{start})	1,240	1,100	2019	1,350	1,450	2020	1,420	2,100	2021	1,480	4,200	2022	1,510	8,900	2023	1,550	14,300	2024 (t_{end})	1,590	22,100
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(2) Change-Point Pattern (P2)																					
Mathematical Definition (M2) Let $\bar{X}(t)$ be a time series. A change point τ is identified if there is a significant change in the trend’s slope β, where: $\beta_2 - \beta_1 \gg 0$ given β_1 is the slope for $t < \tau$ and β_2 is the slope for $t > \tau$.	Textual Description (T2) This pattern identifies a specific point in time like a date, year, etc. (a “turning point”) where a trend’s direction or speed changes dramatically.																				
Instances in Report (E2.text) “There has been a big surge in patent activity beginning in 2017, with an average annual growth of around 45% since then. This coincides with the introduction of transformers in 2017.”	Instances in Data (E2.tab) <table><tr><th colspan="2">Change Point = 2018</th></tr><tr><th>Grant Year</th><th>GenAI Count</th></tr><tr><td>2013</td><td>3,201</td></tr><tr><td>2014</td><td>3,475</td></tr><tr><td>2015</td><td>3,893</td></tr><tr><td>2016</td><td>4,686</td></tr><tr><td>2017</td><td>5,250</td></tr><tr><td>2018</td><td>5,582</td></tr><tr><td>2019</td><td>8,538</td></tr><tr><td>2020</td><td>11,399</td></tr></table>	Change Point = 2018		Grant Year	GenAI Count	2013	3,201	2014	3,475	2015	3,893	2016	4,686	2017	5,250	2018	5,582	2019	8,538	2020	11,399
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